

Task - 7: Web Application Vulnerability Testing

Internship Task – Elevate Labs

This task focused on understanding common web application vulnerabilities and learning how to identify them using security testing tools. The objective was to gain practical knowledge of web application security testing.

1. Introduction to Web Application Security

Web applications are frequent targets of cyber attacks because they are publicly accessible. Testing helps discover weaknesses before attackers exploit them.

2. OWASP Top 10 Overview

OWASP Top 10 is a list of the most critical web application security risks. It helps developers and testers focus on the most dangerous vulnerabilities.

3. Vulnerable Application Setup

A vulnerable application such as DVWA or OWASP Juice Shop was used for testing. This environment allows safe learning and experimentation.

4. Burp Suite Usage

Burp Suite Community Edition was used to intercept HTTP requests and responses. It helps analyze how user input is processed by the server.

5. SQL Injection Testing

SQL Injection vulnerabilities were tested by modifying input parameters. Successful manipulation confirmed insecure database query handling.

6. Cross-Site Scripting (XSS) Testing

XSS was tested by injecting scripts into input fields. This showed how attackers can execute malicious code in browsers.

7. Response Analysis

Application responses were analyzed to identify abnormal behavior and errors. These indicators help confirm vulnerabilities.

8. Mitigation Techniques

- Input validation and sanitization - Use of prepared SQL statements - Proper output encoding - Secure coding practices

Learning Outcome

This task improved my understanding of web application vulnerabilities and testing methods. I learned how SQL Injection and XSS attacks work and how they can be prevented.